

Complex Microstructural identification in metallic alloys using computer vision, particularly Deep Learning (CNNs)

Problem & Approach

The Problem

Manual microstructure analysis is inherently **slow, subjective, and difficult to scale**. Expert interpretation of SEM images introduces variability and limits throughput in materials research and quality control pipelines.

Objective

Automate both **classification** and **feature extraction** of Ti-6Al-4V alloy microstructures — enabling reproducible, high-throughput characterization without manual intervention.

Two-Stage Deep Learning Pipeline

1

Stage 1 — CNN Classification

Convolutional Neural Network identifies and classifies microstructural phases from raw SEM input images across 3 distinct classes (equiaxed, duplex, and lamellar).

2

Stage 2 — Segmentation

Watershed algorithm segments **duplex** microstructures; HOG feature descriptors characterize **lamellar** structures for quantitative analysis.

Methodology

Dataset

Ti-6Al-4V alloy SEM images across **3 microstructural classes**: equiaxed, duplex, and lamellar. Images preprocessed and augmented for robust training.

CNN Architecture

3 convolutional layers with ReLU activations, max-pooling, fully connected layers. Trained with Adam optimizer and Softmax output for multi-class classification.

Convolution Operation:

$$C(i, j) = \sum_m \sum_n I(i + m, j + n) F(m, n)$$

ReLU Activation:

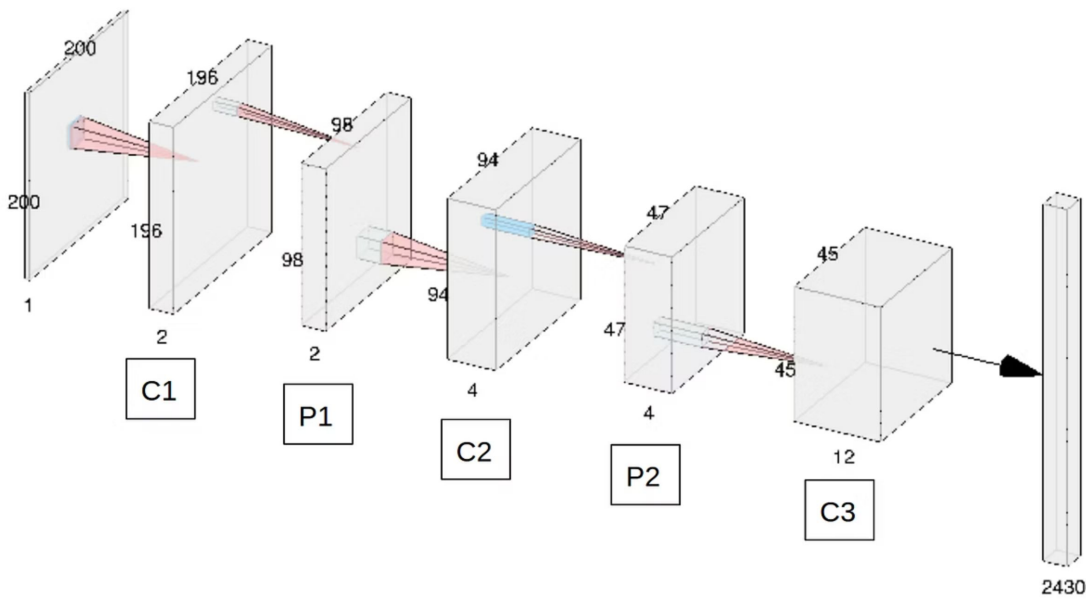
$$f(x) = \max(0, x)$$

Softmax Function:

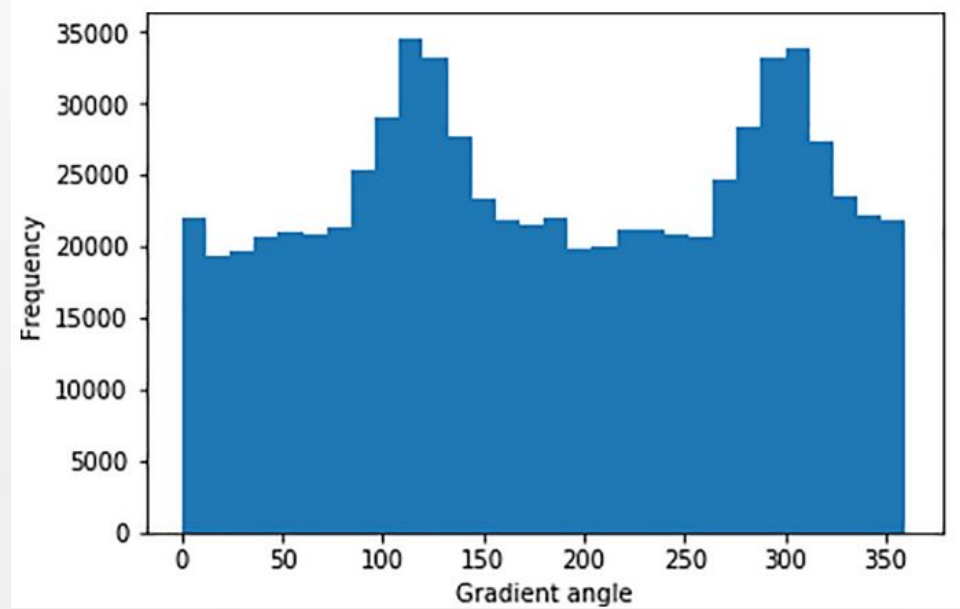
$$p_i = \frac{e^{a_i}}{\sum_k e^{a_k}}$$

Training HyperParameters

Parameter	Value
Learning Rate	0.001
Batch Size	32
Max Iterations / Epochs	Epochs = 50, Patience = 50
Optimizer	Adam
Regularization	L1 ($\lambda = 0.001$)
Activation Functions	Linear (Convolution Layers), Softmax (Output)



HOG



$$G = \sqrt{G_x^2 + G_y^2}$$

Quantification Output:

Phase fraction calculated directly from **pixel ratio** of segmented regions, enabling automated quantitative microstructure analysis without manual measurement.

$$\text{Phase Fraction} = \frac{\text{White pixels}}{\text{Total pixels}} \times 100$$

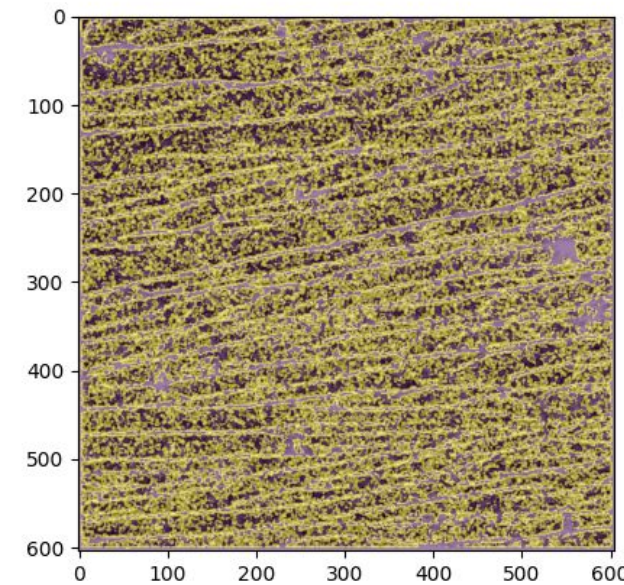
Segmentation

Watershed for duplex

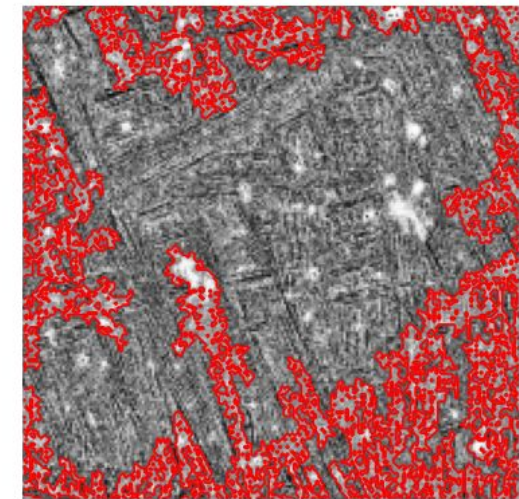
Used to separate equiaxed grains and define boundaries (phase/grain delineation)

HOG for lamellar(Histogram of Oriented Gradients)

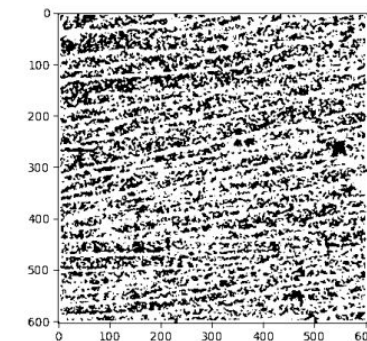
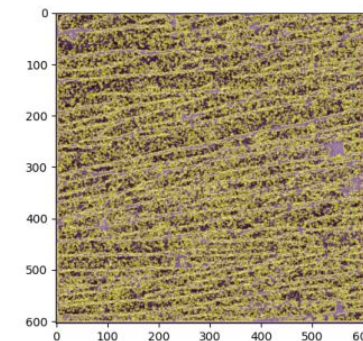
Captures directional/gradient information → useful for lamellar texture analysis



Segmented Lamella image
Original



Segmented Duplex image
Mask



Results & Conclusion

Performance Metrics

94–97%

Classification Accuracy

Across all 3 microstructure classes on held-out test set

Classification Report:				
	precision	recall	f1-score	support
Lamellae	0.97	0.98	0.98	132
Bi-modal	0.97	0.93	0.95	61
Acicular	1.00	1.00	1.00	32
accuracy			0.97	225
macro avg	0.98	0.97	0.98	225
weighted avg	0.97	0.97	0.97	225

Limitations:

a)Dependence on image quality

The model performance is sensitive to SEM image quality (noise, contrast variations, illumination), which can affect both classification and segmentation accuracy.

b)Limited generalization

The model is trained on a specific dataset (Ti-6Al-4V), so it may not perform well on different materials or microstructures without retraining or tuning.

Key Outcomes

- Fully classified and segmented microstructure outputs per input SEM image
- Scalable pipeline — processes large image datasets without expert intervention
- Significant reduction in manual analysis time and inter-analyst variability

